The layer-0 features were free.

The real structure splits by position kind.

Every one of the 23 layer-0 next_node features reported last time fires at pre_color positions, where the current token IS the node, at F1 exactly 1.000. At pre_node positions, where the next node is not yet known, there are ZERO -- in BOTH models.

So job 49729375's pre-registration (a) was UNTESTABLE as written, not false: pooling the two kinds of position mixed a token-identity match into the count. Splitting them also sharpens the main result -- each kind of position carries the DD family it needs, but NOT at the time I predicted. My pre-registration (d) here is falsified.

Same two checkpoints, seed, layers and sparsity target as 49729375, plus --split-by-kind.
The pooled sae_summary.csv came back byte-identical to 49729375 for both arms, so finding 19 reproduces exactly.

job 49744985, COMPLETED 01:00:56, ExitCode 0:0 | commit f24bb40 | scripts/slurm/sudoku2_sae_kindsplit.sbatch

What the model actually sees

Every token is one categorical id into a 34-symbol vocabulary (N=24, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING BEGIN_ROUND SOLVED UNRESOLVED	*section + round markers*
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3	*the q colours*
10-33	N0 N1 ... N23	*one id per node, no structure in the id*

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*
SOLVED	

The graph section is a neighbour table, not an image: node, then its degree neighbours. THE ORDER WITHIN A ROW IS THE LEAK of finding 16.

THIS INSTANCE, 199 TOKENS (seed 50000000032)

HINTS	BEGIN_HINTS N0 COLOR_3 N1 COLOR_3 N2 COLOR_1 N4 COLOR_1 N5 COLOR_2 N9 COLOR_0 ... (33 tokens)
GRAPH	BEGIN_GRAPH N0 N12 N17 N23 N16 N13 N1 N14 N5 N23 N15 N18 ... (145 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N6 COLOR_0 N7 COLOR_1 N8 COLOR_0 N14 COLOR_2 N15 COLOR_2 N18 ... (21 tokens)

Loss is on the REASONING section only: 3 rounds, 8 deductions, 16 target tokens. Everything before BEGIN_REASONING is the problem statement.

One real reasoning trace

16 hints of 24 nodes, then 3 rounds of simultaneous deduction. Hints: N0=3 N1=3 N2=1 N4=1 N5=2 N9=0 N10=2 N11=3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N6 COLOR_0 N7 COLOR_1 N8 COLOR_0 N14 COLOR_2 ...
  6 deduced
BEGIN_ROUND          N23 COLOR_0
  1 deduced
BEGIN_ROUND          N3 COLOR_3
  1 deduced
SOLVED
```

WHY ONE DEDUCTION FIRES

node N6, neighbours [19, 4, 21, 5, 7]

colours known before this round:

```
   [1, 2, 3]  (3 = q-1)
```

only colour left: 0

```
so the trace emits  N6 COLOR_0
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

T=3 rounds, the median of 40 draws from this config (range 1-10); seed 50000000032, shuffle_adj=True; evidence from sudoku_tokens.evidence_rounds (pre-round state, as DD saw it)

Two kinds of position, and why one of them is free

A position is where the model is about to emit a token: pos is the index with tokens[pos + 1] as the target.

THE REASONING SECTION OF THE INSTANCE ON THE PREVIOUS PAGE

BEGIN_ROUND	N6	COLOR_0	N7	COLOR_1	N8	COLOR_0
pos 179	pos 180	pos 181	pos 182	pos 183	pos 184	pos 185
pre_node	**pre_color**	**pre_node**	**pre_color**	**pre_node**	**pre_color**	**pre_node**
-> N6	-> COLOR_0	-> N7	-> COLOR_1	-> N8	-> COLOR_0	-> N14

pre_node (pos = i - 1)

tokens[pos] is BEGIN_ROUND or the PREVIOUS deduction's colour. Nothing at the embedding identifies the node that comes next: the model has to work out which node is deducible.

pre_color (pos = i)

tokens[pos] IS the node whose colour is the target. next_node[v] is therefore matched by a pure token-identity feature, at layer 0, before any attention, with no computation and no leak.

AND THAT IS EXACTLY WHAT LAYER 0 CONTAINS, IN BOTH MODELS:

```
leak-free   pre_color 23 next_node features at best F1 1.000      pre_node 0 features, best F1 0.000
leaked      pre_color 24 next_node features at best F1 1.000      pre_node 0 features, best F1 0.011
```

spin_gnn.sudoku_activations.reasoning_positions; F1 = 1.000 exactly is the signature of a token-identity feature

Layer 0, decomposed

49729375 reported 23 vs 24 interpretable next_node features at layer 0 and I recorded pre-registration (a) as falsified. Here is where all of them were.

model	positions	n	alive	interp.	next_node feats	best F1	median null
leak-free	pre_node	26600	80	11	0	0.000	0.092
leak-free	pre_color	26600	95	30	23	1.000	0.083
leaked	pre_node	26600	85	10	0	0.011	0.096
leaked	pre_color	26600	102	36	24	1.000	0.084

WHAT THIS SETTLES

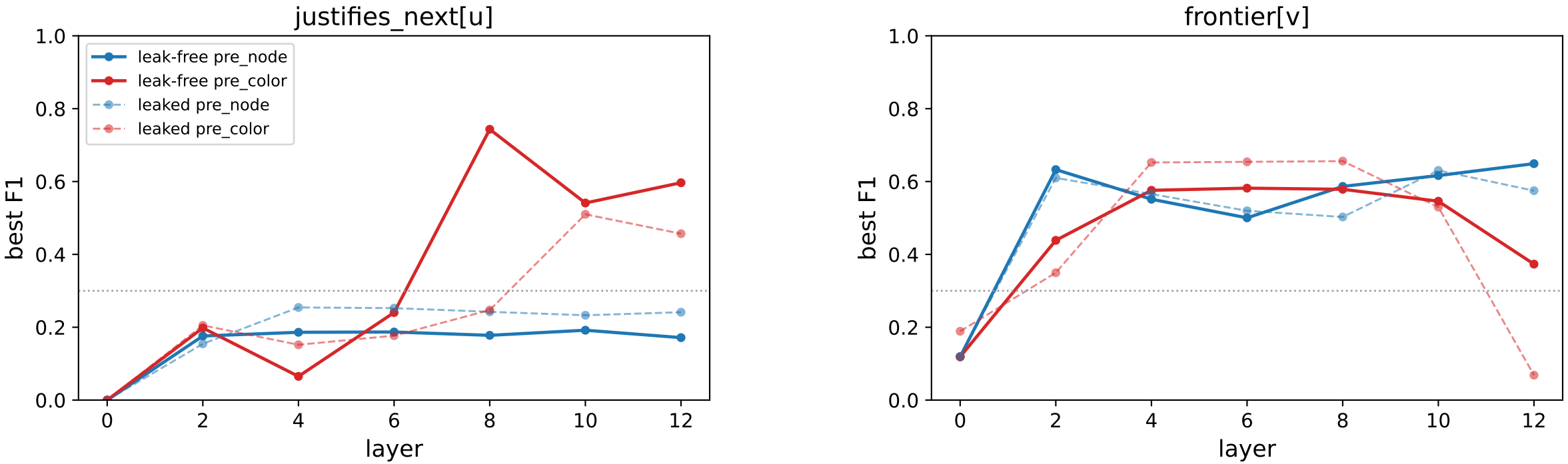
(a) CONFIRMED. The leak-free model has zero next_node-family features above null at pre_node positions and best_next_node_f1 = 0.000, below its own median null (0.092).
(b) CONFIRMED. At pre_color it has 23, at F1 exactly 1.000 -- the token-identity match.
(c) CONFIRMED, in the branch I predicted. The leaked model decomposes identically (0 / 24), so the 23-vs-24 count compared two freebies. 49729375's pre-reg (a) was UNTESTABLE as written and is reclassified from FALSIFIED to UNTESTABLE.

Layer 0 is the token+position embedding, and the only layer in either arm that misses the sparsity target (L0 5.9 / 7.0 against 64): a 2-sparse embedding is what the dictionary finds because that is what is there. Both arms keep ~10 interpretable pre_node features, none next_node.

sae_summary_by_kind.csv, layer 0, target L0=64, 26600 positions per mask

Each kind of position carries a different DD family

Left: the edges that force the colour. Right: the frontier, which selects the node. Both are best F1 among alive features.



THE EDGES ARE REAL AND THE CONTRAST HOLDS -- BUT ONLY AT pre_color.

At pre_color the leak-free model has 69 justifies_next features against the leaked model's 15, and best F1 0.743 vs 0.247 at layer 8 -- finding 19's result, intact and larger than the pooled version.
At pre_node BOTH arms have zero, at F1 0.15-0.25, under the cut. The frontier is the mirror image: the leak-free model has more frontier features at pre_node than at pre_color at every layer above 0 (6 of 6).

Read positively, this is relevant edges at the relevant time -- the known neighbours matter when the COLOUR is being chosen, and the node is already fixed by then. I predicted the other time, and said so in advance.

solid = leak-free, faint dashed = leaked; blue = pre_node, red = pre_color; dotted = the F1>=0.3 interpretability cut

Pre-registrations, as resolved

Written into scripts/slurm/sudoku2_sae_kindsplit.sbatch before the job ran; commit f24bb40.

	claim	verdict	evidence
(a)	leak-free L0 pre_node has ZERO next_node features	CONFIRMED	0 features, best F1 0.000 vs null 0.092
(b)	leak-free L0 pre_color has many, at high F1	CONFIRMED	23 features, best F1 1.000 (exactly 1)
(c)	if both hold in both arms, 49729375's (a) was untestable rather than false	CONFIRMED	leaked decomposes 0 / 24 identically; (a) reclassified
(d)	>=1 justifies_next feature survives at pre_node (L8 or L12), and leak-free beats leaked there	FALSIFIED	0 features at pre_node at every layer, both arms; and leak-free is LOWER there (0.171 vs 0.241 at L12)
(e)	frontier F1 higher at pre_node than pre_color (leak-free)	SPLIT	4 of 6 layers on F1, but 6 of 6 on feature counts (31/31/26/26/35/33 vs 13/14/19/15/14/2)

(d) is the one that cost something, and it is worth being exact about what it cost.

I predicted the edge features would appear while the model is CHOOSING the node, on the theory that the justifying neighbours are what makes a node deducible. They appear only while it is choosing the COLOUR. Both are 'relevant edges at a relevant time', and the second is the one DD's firing rule actually needs -- but it is the weaker of the two claims, because at pre_color the node is already in the context window.

my record on mechanism predictions in this project: 4 confirmed, 1 falsified, 1 split, over the last two jobs

What this changes, and what it does not

CHANGED IN FINDING 19

- Pre-reg (a) goes from FALSIFIED to UNTESTABLE: the 23-vs-24 layer-0 count compared two token-identity freebies and could not have separated the arms.
- The edge result is restated: the leak-free model's justifies_next features live at pre_color positions only. The leak-free vs leaked contrast is unchanged and in fact larger inside that mask than it was pooled.
- The pooled sae_summary.csv reproduced 49729375 byte-for-byte in both arms.

NEW OPEN PROBLEM, FOUND HERE

The leaked model has MORE next_node features at pre_node than the leak-free one (60 vs 15 at L12, best F1 0.741 vs 0.545) -- it is better at knowing which node comes next before emitting it. Finding 16 measured that round events are emitted in ASCENDING NODE INDEX for 100% of rounds, and shuffle_adj permutes adj[v] only, not that. So a next_node feature at pre_node may track 'the index goes up' rather than deduction, in BOTH arms. Permuting within-round event order is a separate control, never run, and cheap.

CAVEATS

- Each mask keeps 26600 of 53200 positions, so a feature at the 1e-3 aliveness floor fires ~26 times instead of ~52. Counts compare WITHIN a mask, not across masks.
- The dictionaries are the pooled ones, scored twice; this does not ask what a dictionary fitted to pre_node positions alone would look like.
- One seed, one checkpoint per arm. best_f1 is a max over 256 predicates; the rolled null controls the argmax, not the interpretation.
- None of this says either model runs DD, only which DD predicates are readable, and when.